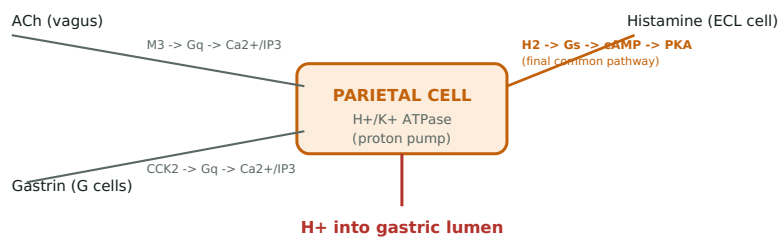


**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which receptor is the **final common pathway** of acid secretion, and which second messenger does it use?
2. Why must PPIs be taken about **1 hour before a meal**?
3. Why should **sucralfate** never be given together with an antacid or an H₂ blocker?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Regulation of acid secretion

Three receptors converge on the parietal cell H⁺/K⁺ ATPase. Histamine/H₂/cAMP is the **final common pathway** — ACh and gastrin also act partly by triggering ECL-cell histamine release.

Receptor	Ligand / source	Pathway
M3	ACh, vagus nerve	G _q → PLC → Ca ²⁺ /IP ₃
CCK2	Gastrin, G cells (antrum/duodenum)	G _q → PLC → Ca ²⁺ /IP ₃
H2	Histamine, ECL cells (fundus)	G _s → adenylate cyclase → cAMP → PKA

Basal Acid Output (BAO), normal adult: **2-5 mEq/hr**. Prostaglandins E₂/I₂ **inhibit** acid secretion (block adenylate cyclase) and **stimulate** mucus + HCO₃⁻ secretion — this is the target of misoprostol.

2 Peptic ulcer disease

WHO	~10% of Western population; duodenal ulcer 4-5× more common than gastric ulcer.
WHY	NSAID use , H. pylori infection, ↑ HCl secretion, inadequate mucosal defense.
GOALS OF Rx	Relieve pain · heal the ulcer · prevent complications (bleeding, perforation) · prevent relapse.
H. PYLORI Rx	Triple: PPI + (metronidazole or amoxicillin) + clarithromycin. Quadruple: bismuth subsalicylate + metronidazole + tetracycline + PPI. 2-week course → ≥90% eradication.

3 Drug classes for acid suppression

Class	Examples	Mechanism	Key use	Key caution
PPI	Omeprazole, lansoprazole, pantoprazole, rabeprazole, esomeprazole	Prodrug; irreversibly, covalently binds H ⁺ /K ⁺ ATPase	Superior to H ₂ RA for PUD/GERD healing; Zollinger-Ellison	Long-term → ↓ vit B ₁₂ ; take ~1h before food (bioavailability ↓ 50% by food)
H₂ antagonist	Cimetidine, ranitidine, famotidine, nizatidine	Competitive, reversible H ₂ block → ↓ cAMP	PUD; acute stress ulcer (IV); GERD (now 2nd line)	Cimetidine inhibits CYP450 (warfarin, phenytoin, theophylline...); gynecomastia; dose-reduce in renal impairment
Antimuscarinic	Pirenzepine (M ₁ -selective), dicyclomine (M ₁ /M ₂ /M ₃)	Blocks ACh-driven acid stimulation	Refractory PUD, Zollinger-Ellison (add-on)	Dry mouth, constipation, urinary retention, arrhythmia — limits use

Prostaglandin analogue	Misoprostol (synthetic PGE1)	↓ histamine-driven cAMP + ↑ mucus/HCO ₃ ⁻	Prevention of NSAID-induced ulcers (± PPI)	Contraindicated in pregnancy (uterine contractions); diarrhea/nausea; shorter duration (~3h) than H2RA/PPI
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4 Discriminator — PPI vs H2 antagonist

The ONE difference: PPIs **irreversibly** inactivate the pump itself (~18h needed to resynthesize → once-daily, longer, more potent suppression). H2 blockers only **competitively and reversibly** block the histamine arm — the ACh and gastrin routes stay open, so suppression is weaker and shorter.

5 Antacids

Type	Agent	Key point
Systemic	NaHCO ₃	Potent (pH >7); absorbed → systemic alkalosis, Na load (worsens CHF/edema), CO ₂ → belching/perforation risk. Casual heartburn only.
Non-systemic	Mg(OH) ₂	Diarrhea/loose stools; excess → hyporeflexia, hypotension, cardiac arrest.
Non-systemic	Al(OH) ₃	Constipation; binds phosphate → hypophosphatemia; toxicity (encephalopathy, osteoporosis) if renal failure.
Non-systemic	Mg(OH) ₂ + Al(OH) ₃	Combined → bowel effects cancel out.
Non-systemic	CaCO ₃	Potent, fast; CO ₂ → distention. Avoid in HTN, heart failure, renal disease.

Drug interactions: antacids alkalinize gastric contents and alter bioavailability of iron, tetracyclines, digoxin, prednisolone, propranolol, quinidine, antimuscarinics; their ions chelate quinolones and tetracyclines. Space doses apart.

6 Mucosal protective (cytoprotective) agents

Agent	Mechanism	Key point
Sucralfate	Sulfated Al-sucrose → viscous paste binds ulcer base for up to 6h; ↑ prostaglandin & HCO ₃ ⁻	Needs acidic pH to activate → never combine with antacids/H2RA. Heals duodenal ulcers only, not gastric; doesn't prevent NSAID ulcers.
Bismuth subsalicylate	Coats ulcer base, inhibits pepsin, ↑ mucus; part of quadruple therapy	Nausea/vomiting; black tongue and stools.

P Must-remember pearls

- H2 receptor = the **final common pathway** of acid secretion (Gs → cAMP) — PPIs act downstream and irreversibly, which is why they win.
- PPIs need an **acidic canaliculus** to activate — timing (1h before a meal) and food matter for efficacy.
- Misoprostol** is contraindicated in pregnancy; **sucralfate** needs acid, so never co-prescribe with antacids or H2RA.
- Cimetidine** is the "interacting" H2 blocker (CYP450 inhibitor) — ranitidine/famotidine are safer.
- H. pylori eradication** = triple (PPI + 2 antibiotics) or quadruple (bismuth + metronidazole + tetracycline + PPI) × 2 weeks, ≥90% success.

Q Test yourself

1. A 45-year-old woman on long-term omeprazole for GERD presents with fatigue and macrocytic anemia. Best explanation?

- A. PPI-induced folate deficiency
 B. PPI-induced B12 deficiency (acid is needed to release B12 from food)
 C. PPI-induced iron deficiency
 D. Unrelated to the PPI

2. Which drug is contraindicated in a pregnant woman who needs NSAID-ulcer prophylaxis?

- A. Sucralfate
 B. Misoprostol
 C. Ranitidine
 D. Omeprazole

3. A patient on cimetidine develops a rising INR while on warfarin. Mechanism?

- A. Cimetidine induces CYP450
- B. Cimetidine inhibits CYP450
- C. Cimetidine displaces warfarin from albumin
- D. Cimetidine increases GI absorption of warfarin

4. Why should sucralfate not be co-administered with an H2 blocker?

- A. Both cause diarrhea
- B. Sucralfate needs an acidic pH to activate, which H2 blockers reduce
- C. They chelate each other
- D. Both compete for renal clearance

5. A duodenal ulcer patient is prescribed quadruple therapy. Which combination is correct?

- A. PPI + amoxicillin + clarithromycin
- B. Bismuth subsalicylate + metronidazole + tetracycline + PPI
- C. H2RA + misoprostol + sucralfate
- D. NaHCO₃ + antacid + antimuscarinic

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. — B — acid is required to release vitamin B12 from food; long-term acid suppression can cause subnormal B12.
- 2. — B — misoprostol is contraindicated in pregnancy because it causes uterine contractions.
- 3. — B — cimetidine inhibits cytochrome P450, potentiating warfarin (and phenytoin, theophylline, etc.).
- 4. — B — sucralfate requires an acidic environment to form its protective paste.
- 5. — B — quadruple therapy = bismuth subsalicylate + metronidazole + tetracycline + PPI.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · from Dr. Begard O. Berzinji's GI Pharmacology lecture (acid-related disorders)
Study alongside the lecture, not instead of it.